

Chapter 1 Fundamentals

Chapter 2 De Rham Cohomology, and Harmonic Differential Form

2.1 The Laplace Operator

Review:

- The scalar product

$$\langle -, - \rangle : \wedge^p V \times \wedge^p V \rightarrow \mathbb{R}$$

$$(v_1 \wedge \cdots \wedge v_p, w_1 \wedge \cdots \wedge w_p) = \det(\langle v_i, w_j \rangle)$$

- If $e_1, \dots, e_d$ is an orthonormal basis, then

$$e_{i_1} \wedge \cdots \wedge e_{i_p}; 1 \leq i_1 < \cdots < i_p \leq d$$

is an orthonormal basis of $\wedge^p V$;

- Given an orientation, then the **star operator**

$$* : \wedge^p(V) \rightarrow \wedge^{d-p}(V); 0 \leq p \leq d$$

$$(e_{i_1} \wedge \cdots \wedge e_{i_p}) \mapsto e_{j_1} \wedge \cdots \wedge e_{j_{d-p}}$$

where $\{j_1, \dots, j_{d-p}, i_1, \dots, i_p\} = \{1, \dots, d\}$.

In particular:

$$*(1) = e_1 \wedge \cdots \wedge e_d$$

$$*(e_1 \wedge \cdots \wedge e_d) = 1$$

$$*(Af_1 \wedge \cdots \wedge Af_p) = \det(A) * (f_1 \wedge \cdots \wedge f_p)$$

- $*(Af_1 \wedge \cdots \wedge Af_p) = \det(A) * (f_1 \wedge \cdots \wedge f_p)$.
- We have:

$$** = (-1)^{p(d-p)} : \wedge^p(V) \rightarrow \wedge^p(V)$$

$$\langle v, w \rangle = *(w \wedge *v) = *(v \wedge *w)$$

$$\wedge(V) = \bigoplus \wedge^p(V), \wedge^p(V) \perp \wedge^q(V), p \neq q$$

$$*(1) = \frac{1}{\sqrt{\det(\langle v_i, v_j \rangle)}}$$

We obtain the operator

$$* : \wedge^p(T_x^* M) \rightarrow \wedge^{d-p}(T_x^* M)$$

i.e. a base point preserving operator

$$* : \Omega^p(M) \rightarrow \Omega^{d-p}(M) \quad \Omega^k(M) := \Gamma(\wedge^k(M)) = \Gamma(\wedge^k(TM))$$

Recall the metric on $T_x^* M$ is given by

$$(g^{ij}(x)) = (g_{ij}(x))^{-1}$$

Therefore,

$$*(1) = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^d$$

Definition 2.1.0.1 Such form $*$ (1) is called **volume form**.

Inner Product

Definition 2.1.0.2 Suppose $\alpha, \beta \in \Omega^p(M)$. Then, define the L^2 -product of $\Omega^p(M)$ as following:

$$\begin{aligned} (\alpha, \beta) &= \int_M \langle \alpha, \beta \rangle * (1) \\ &= \int_M \det(\langle \alpha_i, \beta_j \rangle) \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^d \\ &= \int_M \langle \alpha, * \beta \rangle \end{aligned}$$

The product on $\Omega^p(M)$ is obviously bilinear and positive definite.

Operator d^*

Definition 2.1.0.3 Consider the operator $d : \Omega^p M \rightarrow \Omega^{p+1} M$. Then, its adjoint d^* on $\oplus \Omega^p(M)$,

$$(d\alpha, \beta) = (\alpha, d^* \beta)$$

Therefore, $d^* : \Omega^p M \rightarrow \Omega^{p-1} M$.

◇ **Lemma 2.1.0.4** Let $d^* : \Omega^p(M) \rightarrow \Omega^{p-1}(M)$ satisfies:

$$d^*(-) := (-1)^{d(p+1)+1} * [d(*(-))]$$

◦ **Proof ...**

□

2.1.1 Laplace-Beltrami Operator

Definition 2.1.1.1 The **Laplace-Beltrami operator** on $\Omega^p(M)$ is

$$\Delta = dd^* + d^*d : \Omega^p(M) \rightarrow \Omega^p(M)$$

And

Definition 2.1.1.2 $\omega \in \Omega^p(M)$ is called **harmonic** if

$$\Delta\omega = 0$$

★ **Corollary 2.1.1.3** Δ is self-adjoint. i.e.

$$(\Delta\alpha, \beta) = (\alpha, \Delta\beta)$$

◦ **Proof ...**

□

◇ **Lemma 2.1.1.4** (Harmonic Form) $\Delta\alpha = 0 \Leftrightarrow d\alpha = 0, d^*\alpha = 0$.

◦ Proof ...

□

★ Corollary 2.1.1.5 ...

◦ Proof ...

□

◇ Lemma 2.1.1.6 ...

◦ Proof ...

□

Local Representation

◇ Lemma 2.1.1.7 ...

◦ Proof ...

□

♠ Proposition 2.1.1.8 For a differentiable $f : M \rightarrow \mathbb{R}$,

$$\Delta f = -\frac{1}{\sqrt{g}} \frac{\partial}{\partial x^j} (\sqrt{g} g^{ij} \frac{\partial f}{\partial x^i}) \quad g = \det(g_{ij})$$

◦ Proof ...

□

2.2 Representing Cohomology Classes by Harmonic Forms

Definition 2.2.0.1 ...

► **Theorem** (Existence of Harmonic Form in Homology Class)

◦ Proof ...

□

2.3 Generalization: Sobolev Space, Elliptic

2.3.1 Sobolev Space by Bundles

Definition 2.3.1.1 Let E, F be vector bundles over compact, oriented, differentiable manifold M . Let $\Gamma(E), \Gamma(F)$ be the spaces of differentiable sections.

Sobolev spaces of sections can be defined with the help of bundle charts: Let (f, U) be a bundle chart of E . f then identifies $E|U$ with $U \times \mathbb{R}^n$. A section s of E is then contained in the Sobolev space $H^{k,p}(E)$, if for any such

bundle chart, and $U' \subseteq U$ we have

$$p_2 \circ f \circ s|_{U'} \in H^{k,p}(U', \mathbb{R}^n)$$

where p_2 is the projection $p_2 : U' \times \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Differential Operator

Definition 2.3.1.2 A linear map

$$L : \Gamma(E) \rightarrow \Gamma(F)$$

is called **(linear) differential operator** of order l from E to F

In bundle chart, we write L as

$$L = P_0(D) + \dots + P_l(D)$$

Elliptic Operator

Definition 2.3.1.3 ...

2.4 Some of Unclassified Conclusions

♠ **Proposition 2.4.0.1** Let $\omega \in \Omega^1(S^2)$ be a 1-form of S^2 , suppose that

$$\varphi^* \omega = \omega$$

$\varphi \in SO(3)$. Then,

◦ **Proof** ...

□

♠ **Proposition 2.4.0.2** ...

◦ **Proof** ...

□

Chapter 3 Parallel Transport, Connections and Covariant Derivatives

Some of reviews in Riemannian geometry.

3.1 title

3.2 Metric Connections, The Yang-Mills Functional

3.3 The Levi-Civita Connection

3.4 Connections for Spin Structures, and the Dirac Operator

Chapter 4 Geodesics and Jacobi Fields

Chapter 5 Symmetric Spaces, and Kahler Manifolds

Chapter 6 Morse Theory, and Floer Homology

Chapter 7 Variational Problems from Quantum Field Theory

Chapter 8 Harmonic Maps